



BINOMIAL DISTRIBUTION

FAST MCQ TRICKS — BY THUNDERSTUDY

Think of a Coin Game: Same experiment repeated, only 2 outcomes each time (Head/Tail, Success/Failure, Pass/Fail).

✓ STEP 1: 4-Second Identification Trick

Check Questions	Condition Met?
Fixed number of trials?	✓
Only 2 outcomes?	✓
Probability same every time?	✓
Trials independent?	✓

All YES = Use Binomial Formula!

⚡ STEP 2: The Magic Formula

$$P(X=x) = {}^n C_x \cdot p^x \cdot q^{(n-x)}$$

Memory Trick: **"Choose × Success × Failure"**

n = total trials | x = success needed | p = success prob | $q = 1 - p$



STEP 3: SUPER FAST MCQ SHORTCUTS

★ **Trick 1 (At least one):** $P(\text{at least one}) = 1 - P(\text{none})$. Always faster than summing cases!

★ **Trick 2 (Mean/Var):** Mean = np | Variance = npq | SD = $\sqrt{npq}$.

★ **Trick 3 (Symmetry):** If $p = 0.5$, then $P(X=x) = P(X=n-x)$. (Example: $P(2) = P(4)$ if $n=6$).

★ **Trick 4 (Most Probable - Mode):** Mode = $(n+1)p$. Take the nearest integer.

STEP 4: 10-Second MCQ Strategy

1. Identify **n, p, q, x**.
2. Look for keywords: At least (Use 1-none), Average (Use np).
3. Use Symmetry (If $p=0.5$).
4. Apply formula only if no shortcut fits.

Example (Solved in 15 sec)

Coin tossed 3 times. Find probability of exactly 2 heads.

$n=3, p=0.5, q=0.5, x=2$

$$P = {}^3C_2 (0.5)^2(0.5)^1 = 3 \times (0.5)^3 = 3/8$$

Done.

ULTIMATE MEMORY LINE

"Binomial = Fixed trials + Same prob + Two outcomes"
"At least $\rightarrow$ 1 - none"

By Thunderstudy